nature portfolio

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No software was used for data collection.
Data analysis	The method developed in this study has been made available as an R package and can be installed from: https://github.com/aalto-ics-kepaco/survivalfm. The scripts used for conducting the analyses presented in this manuscript are available on GitHub: https://github.com/aalto-ics-kepaco/survivalfm-analysis. Data analyses were conducted using R statistical software, version 4.3.1. Other required R packages: - glmnet (version 4.1.8) - survival (version 3.8.3) - survMisc (version 0.5.6) - nricens (version 1.6)

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

This study used data from the UK Biobank cohort, accessed via application ID 147811. The UK Biobank study was approved by the North West Multi-Centre Research Ethics Committee and all participants provided written informed consent.

Individual-level UK Biobank data are available under restricted access due to participant confidentiality and data privacy regulations. Researchers can apply for access through the UK Biobank Access Management System: https://www.ukbiobank.ac.uk/enable-your-research/apply-for-access. Access is granted to bona fide researchers for health-related research, following a formal application and approval process.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	Self-reported sex (UK Biobank Field ID: 31) was included as a covariate in the models developed in this study and therefore included in all analyses. The method developed in this study inherently accounts for interactions among all included covariates, including sex and its potential interactions with other variables. Sex was not considered in the study design, as the research question is not expected to yield differential results between sexes. Therefore, a combined analysis was deemed more appropriate for addressing the study's objectives. The findings from this study are applicable to both sexes.
Reporting on race, ethnicity, or other socially relevant groupings	The UK Biobank dataset includes ethnicity information (UK Biobank Field ID: 21000) for almost all participants. This was included as a covariate in the models developed in this study to ensure applicability to diverse populations.
Population characteristics	Characteristics of the study population are detailed in Supplementary Tables 1 and 6.
Recruitment	UK Biobank has recruited approximately 500,000 participants aged between 40-69 years in 2006-2010 from across the UK. Recruitment of study participants in the UK Biobank is based on volunteer enrollment. Participants are therefore on average healthier than the general population.
Ethics oversight	The UK Biobank study was approved by the North West Multi-Centre Research Ethics Committee and all participants provided written informed consent. The study protocol is available online (https://www.ukbiobank.ac.uk).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	The study relied on data from the UK Biobank. No specific procedure to determine the sample size a priori was applied, as the study used all available data points in each data modality. Participants recruited from England and Wales (93% of the cohort) were included in the training dataset, while participants recruited from Scotland (7% of the cohort) were included in the test dataset. Sample sizes in the training and test sets across each data modality are detailed in Supplementary Tables 1 and 6.
Data exclusions	For each analyzed disease, individuals with the same prevalent disease at the study baseline were excluded in order to remove samples with an already existing clinical presentation of the disease.
Replication	The models derived from the training set (participants recruited in England and Wales) were replicated by independently testing them once in the subcohort of participants recruited in Scotland.
Randomization	This is an observational cohort study and thus randomization is not relevant.
Blinding	This is an observational cohort study and thus blinding is not relevant.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	Not applicable.
Novel plant genotypes	Not applicable.
Authentication	Not applicable.